

Jiayi Zhang

Master of Data Science

Thompson Rivers University, Kamloops, British Columbia, Canada

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Research Interests

My research interests lie at the intersection of statistics and data science, with a particular focus on incomplete data analysis, effective sample size estimation, survival analysis, statistical inference, and computational statistics. My undergraduate training in mathematics motivated me to apply quantitative and statistical methods to real-world data problems. During a data science group project, I became interested in how missing, truncated, and censored observations reduce the amount of information contained in a dataset. In such settings, the nominal number of observations may not accurately represent the effective amount of statistical information available for estimation and inference. This observation inspired my current thesis research, which develops and evaluates methods for measuring information loss and estimating effective sample size for incomplete data.

Education

Thompson Rivers University

Master of Data Science, Thesis-Based Program

September 2024 – August 2026 (Expected)

Kamloops, British Columbia, Canada

- Thesis supervisor: Professor Mateen Shaikh.
- Research focus: Effective sample size estimation for incomplete, truncated, and right-censored continuous data.
- Thesis Title: Augmented Sample Size Calculation With Truncated Data
- Relevant Selected coursework: Linear Models, Theoretical Machine Learning, Database, Data Science Experimental Design.

Beijing University of Technology

Bachelor of Science in Information and Computing Sciences

September 2020 – July 2024

Beijing, China

- Developed a strong foundation in mathematical analysis, algebra, probability, statistics, numerical analysis, optimization, differential equations, and mathematical modeling (MATLAB), mathematical programming (R, Python).

Research Accomplishments to Date

- Formulated a research framework for distinguishing nominal sample size from effective sample size when observations are incomplete, truncated, or right-censored contained over 500 rows of core coding.
- Designed several probability-based weighting strategies to quantify the information contribution of incomplete observations and compare with the traditional informative methods.
- Constructed simulation experiments to compare the proposed strategies under different sample sizes, truncation proportions, censoring proportions, effect sizes, and variance settings.
- Evaluated the methods using statistical criteria including bias, mean squared error, confidence interval coverage, Type I error, power, and standardized test-statistic distributions.
- Applied the proposed methodology to several real datasets, testifying the performance of methods in truncated or right censored data. .
- Completed initial body drafts of the literature review, methodology, simulation results, introduction, and discussion sections of the thesis.
- Revised the graphical presentation of simulation findings to improve the interpretation and comparison of the proposed strategies.

Thesis Progress Timeline

September 2025	Started the thesis project and had the first in-person campus meeting with Professor Mateen Shaikh.
October 2025	Conducted background research and reviewed relevant literature. Finalized the thesis topic as effective sample size re-estimation for incomplete data.
November 2025	Developed the initial implementation in Jupyter Notebook and tested the algorithm on a simplified dataset to assess its computational feasibility.
December 2025	Organized and synthesized the literature reviewed during the previous term and began drafting the thesis methodology section.
January 2026	Continued writing the literature review and obtained preliminary results from simulated datasets.
February–March 2026	Conducted extensive simulation experiments, analyzed preliminary findings, and drafted the results section.
April 2026	Revised the plotting and visualization methods used in the results section. Investigated unstable simulation results through additional debugging, sensitivity analysis, and methodological review.
May 2026	Drafted the introduction and discussion sections and incorporated the main empirical findings into the thesis.
June 2026	Continued refining the methodology, real-data analysis, comparative effective sample size estimators, and interpretation of simulation results.

Awards and Academic Recognition

- Dr. Sherman Jen Graduate Entrance Award, Thompson Rivers University.
- Three semesters Thesis Scholarship awarded in recognition of strong academic and research performance.
- Research Assistant 2025 Sep - Now
- Participated in Three Minutes Thesis speech competition for social networking

Languages

Chinese (native), English (professional working proficiency; IELTS 7), Japanese (JLPT N2), and French (intermediate proficiency).